DOC-24 | CONTROLLED SOURCE SYNCHRONIZATION

Cross-reference-only revision notice

v1.0-RC4 | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

Control	Selected value
Revision	v1.0-RC3 -> **v1.0-RC4**
Semantic change	**NONE.** This revision changes source version/count/path references only.
Wave-2 semantic authorities	DOC-07 RC2 | DOC-08 RC2 | DOC-15 RC8 | DOC-16 RC5
Canonical HTTP companion	contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml
OpenAPI SHA-256	bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00
RC8 estate	113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL

DOCUMENT-SPECIFIC SYNCHRONIZATION

• Deployment architecture is unchanged. CI/CD contract gates now pin DOC-15 v1.0-RC8 and DOC-16 v1.0-RC5.

• Canonical OpenAPI path: contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml. SHA-256: bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00.

• Contract synchronization gate: 113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs; 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL. Production build/promotion still uses the same immutable-artifact and no-code-first rules.

SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC7/old HTTP: inherited printed pages 1, 4, 11, 12, 13, 24, 25, 27, 28, 30; RC4: inherited printed pages 1, 4, 24.

IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.

BLOG-DOC-24

Environment, Docker, CI/CD, Database Migration, Deployment, Release & Rollback Contract

Production delivery authority for the Hybrid Monorepo: environment separation, validated configuration, Docker images, YAML deployment sources, local Compose, CI/CD, affected builds, database migration/seed safety, staged promotion, health gates, immutable artifacts, rollback and production release recovery.

4	17	8	1
promotion tiers	initial application packages	QA21 release gates	immutable release artifact set

CORE RELEASE RULE	Build once, verify once, promote the same immutable artifact digests. Production never receives a source rebuild, ad hoc schema mutation, hand-edited manifest, unverified secret, tenant-blind configuration or database change that makes the previous compatible application impossible to restore safely.

Document control	Value
Status	REVIEW CANDIDATE - DOC-23 RC2 synchronized; controlled DOC-20 backup release-gate correction applied; freeze after DOC-24 approval
Version	1.0-RC2
Date	10 September 2026
Upstream baseline	DOC-01..14 RC1; DOC-15 RC7 + versioned OpenAPI; DOC-16 RC4; DOC-17 RC2; DOC-18 RC2; DOC-19 RC2; DOC-20 RC1; DOC-21 RC1; DOC-22 RC1; DOC-23 RC2
Primary authority	Environment tiers; configuration/secret injection; Docker/Dockerfile rules; YAML/deployment source conventions; local Compose/selective startup; CI/CD and affected graph; immutable artifacts; DB migration/seed execution; staging/production gates; provider wiring; health probes; rollout/promotion/rollback; production migration safety.
Explicit non-authority	Does not redefine product features, 14-site ownership, permissions, Page IDs, API operations/schemas, DOC-08 database semantics, DOC-18 jobs/Redis keys, DOC-17 media semantics, DOC-19 security policy, DOC-20 SLO/RPO/RTO, DOC-21 expected QA behavior or DOC-23 codebase architecture.
Implementation evidence	Deployment source/runtime evidence was not supplied. DOC-22 implementation state remains NOT_VERIFIED / EVIDENCE_REQUIRED until a candidate pipeline, images, migrations, manifests and QA evidence set proves this contract.
Supersedes	BLOG-DOC-24 v1.0-RC1 implementation-runtime references only. Environment, deployment, migration, release, rollback, QA, SiteContext, security, SLO and provider semantics remain preserved except where RC2 explicitly aligns implementation wiring to DOC-23 RC2.

DEVELOPER LOCK	A developer must not copy production secrets into .env, run schema push from application startup, rebuild between staging and production, deploy every site for a one-site change, use a mutable image tag as release identity, skip QA21 because CI is green, auto-run destructive down migrations, or make rollback depend on restoring an incompatible database without an approved recovery decision.

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Section	Subject
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ONE SOURCE OF TRUTH	DOC-24 owns deployment execution and provider wiring only. It implements the DOC-23 repository and stack, the DOC-20 operational targets and the DOC-21 release gates. It never changes an upstream business, tenant, schema, API, security, worker or QA meaning to fit a deployment tool.

1. Purpose, Authority and Conflict Rules

BLOG-DOC-24 converts the frozen Hybrid Monorepo and operational requirements into one reproducible path from a developer commit to Development, Test, Staging and Production. It also defines how the release is stopped, rolled back or recovered when a database change is not safely reversible.

- Environment separation is a security and evidence boundary, not a naming convention.
- Every release is identified by one Git commit plus immutable OCI/image/artifact digests and a release manifest.
- Database schema changes occur only through DOC-08-approved ordered raw SQL migrations executed by deployment tooling.
- QA21 remains the release-blocking test authority. DOC-24 schedules gates; it does not redefine expected results.
- Provider defaults are subordinate to SiteContext, RLS, no-store, queue, media, security and SLO contracts.

CONFLICT RULE	If deployment would require a new API operation, DB table/column, permission, job type, security bypass, shared site runtime, relaxed SLO, weakened RLS or altered public route behavior, deployment stops and the owning upstream authority is reopened. DOC-24 cannot solve product/contract gaps with infrastructure workarounds.

RC2 RECONCILIATION - This revision synchronizes DOC-24 with DOC-23 RC2: public site runtime is Next.js 16.3 on Node 24; API and Worker are Python 3.13 with FastAPI/Celery; PostgreSQL access is SQLAlchemy/Psycopg while raw SQL remains schema authority; JS/TS uses pnpm-lock.yaml and Python uses uv.lock. No product, API, DB, worker, security, QA, SLO or release semantic is redesigned.

2. Frozen Upstream Baseline and Delivery Invariants

Authority	Frozen input	DOC-24 consequence
DOC-01..14	Topology, tenant isolation, page/application ownership and public/Admin behavior	No merged public runtime; Host/SiteContext preserved end-to-end.
DOC-15 RC7	112 ops / 85 paths / 217 schemas / 1,297 refs	OpenAPI file is exact contract; deployment may not invent health/business endpoints inside /api/v1.
DOC-16 RC4	SEO/search/sitemap/canonical/redirect	Edge/cache/routing cannot change indexability, canonical or redirect semantics.
DOC-17 RC2	S3-compatible media + private original + fixed derivatives	Provider buckets/CDN must keep origin private and public derivatives sanitized.
DOC-18 RC2	6 queues / 11 jobs / Redis keys / dedupe	Broker/cache services must preserve queue durability and separate Redis planes.
DOC-19 RC2	Auth/session/MFA/CSRF/CORS/secrets	Secret injection, security state and edge headers must preserve exact security mechanics.
DOC-20 RC1	Scale/SLO/health/PITR/RPO/RTO	Deployment must meet capacity, readiness, alert and recovery targets.
DOC-21 RC1	180 QA21 IDs + G21-0..7	Release cannot promote without required evidence and 100% applicable PASS.
DOC-22 RC1	Trace/evidence/change propagation	No implementation status is upgraded without immutable evidence.
DOC-23 RC2	Mixed-language Hybrid Monorepo, Next.js public apps, Python API/Worker, two ecosystem locks, revised dependency graph	Deployment consumes the revised implementation stack and exact app/package boundaries; no new codebase architecture.

Invariant	Locked delivery rule
DEP24-INV-01	Same commit/digests move Staging -> Production; no production rebuild.
DEP24-INV-02	A site-local change can deploy only that site; other 13 site images remain untouched unless affected graph says otherwise.
DEP24-INV-03	Application startup never changes database schema.
DEP24-INV-04	Production DB migrations are explicit, serialized and run before code that requires them.
DEP24-INV-05	Previous production application must remain compatible after ordinary expand migrations so application rollback is possible.
DEP24-INV-06	Secrets are injected at runtime from security authority; no secret in image, build args, source, CI artifact or browser bundle.

Authority	Frozen input	DOC-24 consequence
DEP24-INV-07		Runtime environment never changes SiteContext/permission meaning.
DEP24-INV-08		Release promotion uses QA21 evidence tied to exact commit/digests/environment.
DEP24-INV-09		No production rollout proceeds while required backup/PITR/readiness release preflight is failed.
DEP24-INV-10		Future site deployment adds one site app/edge route only; shared API/Worker/DB services are not cloned.

3. Environment Architecture and Promotion Topology

ID	Tier	Execution location	Data rule	Purpose
ENV24-DEV	Development	Developer workstation + infrastructure/local/compose.yaml	Local/dummy data; never production credentials	Source iteration; selective site/API/Admin/Worker startup
ENV24-TEST	Test	Ephemeral CI runner + Testcontainers/Compose	Generated deterministic fixtures	Unit/integration/contract/DB checks; destroyed after run
ENV24-STG	Staging	Separate AWS non-production account/VPC/EKS + managed dependencies	Synthetic/approved non-production data	Production-shaped release candidate; all app classes available
ENV24-PROD	Production	Separate AWS production account/VPC/EKS + managed HA dependencies	Production data	Only promoted immutable artifacts; strict manual gate

QA21 ENV21-UNIT/INT/E2E/LOAD/DR remain test evidence classes, not additional promotion tiers. ENV21-UNIT/INT normally execute in ENV24-TEST. ENV21-E2E executes against the exact Staging release candidate. ENV21-LOAD runs on a dedicated production-shaped load cluster/node group in the non-production account, isolated from normal staging noise. ENV21-DR uses a disposable isolated recovery account/VPC/target and real backup artifacts; it never restores over Production.

```
Commit -> ENV24-TEST -> immutable build/artifact set -> ENV24-STG
       -> G21-0..G21-7 evidence -> manual production approval
       -> ENV24-PROD same digests -> post-deploy smoke/monitor -> release record
```

4. Environment Data and Separation Policy

Boundary	Required separation
Credentials	Unique per environment; production credentials cannot authenticate to non-production resources and vice versa.
Network	Staging/Production use separate accounts/VPCs/clusters/security groups. No flat cross-environment service discovery.
Database	One environment cannot connect to another environment database endpoint. Production restore targets are isolated until promoted through DR procedure.
Redis/Broker	Control/cache and broker Redis remain separate inside each environment; no shared cross-environment Redis.
Search	Distinct index domains/collections and aliases per environment.
Object storage	Distinct buckets/prefix roots and KMS keys per environment; production originals never copied to Development/Test.
Security state	Distinct DynamoDB-compatible table and encryption/key namespace per environment.
Vault/secrets	Separate environment namespaces/policies/Transit keys; no production secret read by non-production workload identity.
Telemetry	Environment is mandatory dimension; access to production telemetry does not grant application/site authority.

NO PROD CLONE	Production data is not a normal Staging seed. If a production-derived dataset is ever required for a controlled

	investigation, it must be separately approved, minimized/redacted under DOC-19 and treated as sensitive evidence; this document does not create such a workflow.

5. Configuration Model and Environment Variables

DOC-23 fixes `BP_<AREA>_<NAME>` naming and startup validation. DOC-24 fixes delivery: non-secret environment values are injected by immutable deployment manifests/ConfigMaps; secret material is mounted or obtained through workload identity/Vault and is never committed. Every service parses configuration once into an immutable typed object before readiness becomes true.

Class	Source	Examples	Rule
STATIC	Committed repository config	package.json, turbo.json, tsconfig, pyproject.toml, Ruff/mypy/Import-Linter and dependency rules	No secrets; versioned with commit.
RUNTIME_NONSECRET	Deployment manifest/ConfigMap	ports, endpoints, resource names, feature-neutral timeouts	Validated at startup; can differ per env without rebuild.
SECRET_REFERENCE	Environment variable points to mounted file or identity	DB/Redis/SMTP secret file path, Vault auth role	Variable is a reference/path, never the secret value.
WORKLOAD_IDENTITY	Kubernetes ServiceAccount/IAM role	S3/DynamoDB/monitoring access where supported	Short-lived provider credentials; no static cloud key.
BROWSER_ALLOWLIST	Generated runtime-config JSON	public API base path, release ID, safe telemetry endpoint	No database/Redis/search/storage/Vault/security-store values.
FAIL CLOSED	Missing, malformed or forbidden required configuration causes startup/readiness failure. Runtime modules never substitute a Development default in Staging/Production and never infer a site from an app package when Host/SiteContext resolution fails.		

6. Secret Injection and Workload Identity

- Production uses an external Vault-compatible HA service with Transit. Staging uses an independently scoped non-production Vault namespace/service. Vault is not embedded into application containers.
- EKS workloads authenticate to Vault through workload identity/Kubernetes auth. Secret material is rendered into memory-backed `/run/secrets/bp/` files or obtained through a short-lived client; it is not baked into ConfigMaps or image layers.
- AWS S3, DynamoDB security-state and telemetry access use per-workload IAM roles/service accounts where supported. Static AWS access keys are prohibited.
- PostgreSQL site-runtime and platform-runtime credentials remain distinct; they are supplied as separate secret references and cannot be merged into one superuser connection string.
- Control Redis and broker Redis use separate endpoints and credentials. A single Redis URL cannot point both roles at the same logical instance in Staging/Production.
- SMTP/SES credentials and invitation/reset delivery secrets remain restricted to auth/mail composition paths.
- Rotation is performed without source/image rebuild; workloads reload through controlled restart or provider-supported rotation while honoring DOC-19 session/token rules.

7. Docker Image Architecture

Image ID	Deploys	Runtime	May contain
IMG24-SITE-*	One image per affected public site app	Node.js 24.x + Next.js 16.3.x SSR	Owning site package + Next.js production output + site-local assets only
IMG24-ADMIN	Central Admin	Unprivileged static web server	Vite build output + generated safe runtime config loader; no server secrets
IMG24-API	API	Python 3.13.x + FastAPI	apps/api + required Python domain/application/adapters + canonical/generated API contract artifacts
IMG24-WORKER	Worker	Python 3.13.x + Celery 5.6.x	apps/worker + Python backend/background/adapters; same image used by six queue profiles + scheduler path
IMG24-MIGRATE	Migration tool image	Python 3.13.x + frozen uv environment + PostgreSQL execution tooling	database/migrations,seeds,verification + one-shot raw-SQL runner; operational artifact, not a product app

- Images are multi-stage and reproducible. JavaScript/TypeScript site/Admin build stages use the authoritative `pnpm-lock.yaml` with frozen pnpm installation. Python API/Worker/migration stages use the authoritative `uv.lock` with frozen uv synchronization. Neither lockfile replaces the other.
- Final runtime images run as non-root UID/GID 10001, use read-only root filesystem where the runtime permits, and write only to an explicit ephemeral `/tmp` volume.
- No `.env`, Git metadata, test fixture secrets, source-only credentials, package-manager cache or SSH material is copied to final layers.
- Base images are pinned by OCI digest in release build configuration even when human examples show a semantic tag.
- Each image carries OCI labels for Git SHA, build ID, application/package ID, source repository and build timestamp. Deployment identity uses image digest, not mutable tag.
- Production containers do not install packages at startup and do not perform database migrations during startup.

8. Dockerfile Standards by Application Class

```
# Dockerfile class contract - illustrative, not a second repository authority

PUBLIC SITE: Node 24 builder/runtime + pnpm-lock.yaml
  - build exactly one Next.js 16.3 site package
  - final image contains only that site's production output/assets

ADMIN: Node 24 builder + pnpm-lock.yaml -> unprivileged static runtime

API: Python 3.13 builder/runtime + pyproject.toml + uv.lock
  - frozen uv dependency sync
  - package-owned FastAPI ASGI launcher
  - no Node backend runtime and no schema push

WORKER: Python 3.13 + same uv.lock
  - package-owned Celery/worker launcher selected by BP_WORKER_PROFILE
  - DOC-18 retry/dedupe/lease/DLQ rules override Celery defaults

MIGRATE: Python 3.13 + frozen uv.lock + PostgreSQL execution tooling
  - one-shot ordered raw DOC-08 SQL/seeds/verification only
  - SQLAlchemy metadata/create_all is never migration authority

All final images run non-root and are promoted by immutable OCI digest.
```

Exact package-owned server entry modules remain application implementation details inside DOC-23 folders. DOC-24 freezes the container stage/output/runtime boundary: Node public runtime is not reused for API/Worker, Python API/Worker images do not absorb public-site presentation code, and the migration image has no API/Worker server entrypoint.

9. YAML Conventions and Deployment Source Layout

```
infrastructure/
  local/
    compose.yaml
    env.example
  images/
    site.Dockerfile
    admin.Dockerfile
    api.Dockerfile
    worker.Dockerfile
    migrate.Dockerfile
  templates/
    kubernetes/
      base/
      overlays/staging/
      overlays/production/
    aws/
      cloudformation/
    ci/
      buildspec-pr.yaml
      buildspec-release.yaml
      buildspec-deploy.yaml
    observability/

tooling/scripts/
  ci/
  db/
  release/
```

Rule	Contract
Extension	`.yaml` only for new deployment/Compose/CI/Kubernetes YAML; do not mix `.yml`.
Encoding	UTF-8, LF, final newline, two-space indentation, no tab indentation.
Keys	lowercase/kebab where provider allows; environment variables retain `BP_*` UPPER_SNAKE_CASE.
Secrets	No literal credential/token/password/private key in YAML. Reference Vault/workload identity only.
Images	Kubernetes Production manifests use image digest (`repo@sha256:...`), not `latest` or mutable tags.
Metadata	Every workload has app/component/environment/release labels; public site workloads also carry diagnostic site/app identity label that is not authorization.
Anchors	Release-critical Production YAML avoids YAML anchors/merge keys so rendered manifest review is explicit.
Validation	All YAML is parsed/linted in CI and Kustomize/Compose rendered output is saved as evidence.

10. Local Compose Dependency Strategy

`infrastructure/local/compose.yaml` is dependency infrastructure, not a second application architecture. Site/Admin source processes use the pnpm workspace; API/Worker use `uv run` through their Python projects, optionally invoked by the Turborepo bridge tasks frozen in DOC-23 RC2. Compose supplies disposable dependency interfaces only.

Service	Profile	Purpose
postgres	core	PostgreSQL 17 compatible, migrations/seeds/verification target
redis-control	core	DOC-18 cache/control/dedupe/lock plane
redis-broker	core	Dedicated Celery/Kombu Redis broker plane with durable/noeviction local settings; includes the DOC-23 broker-private delayed-work mechanism.
opensearch	search	Derived public search engine

Service	Profile	Purpose
minio	media	S3-compatible private development object store
security-state	security	DynamoDB-API-compatible local conditional/TTL store
vault-dev	security	Development-only Vault-compatible secret/Transit interface with dummy keys
smtp-sink	security	Non-delivering local SMTP capture
otel-collector	observability	Optional local trace/metric/log collection

```
# Minimal content/API work
docker compose -f infrastructure/local/compose.yaml --profile core up -d
# Turborepo bridge delegates API dev to uv; Node is orchestration only.
pnpm turbo run dev --filter=./apps/api
pnpm --filter @bp/site-01-whats-your-rashi dev

# Add search/media/security only when needed
docker compose -f infrastructure/local/compose.yaml --profile core --profile search --profile media --profile security
up -d
```

11. Selective Local Application Startup

Developer task	Required startup behavior
One public site	Start required dependency profiles + API + selected site package only. Other 13 site source servers remain stopped.
Admin work	Start core/security as required + API + Admin. Public site apps are optional unless testing Preview/public handoff.
Worker work	Start core + the dependency profile for the worker family + API/Worker as needed. One Worker process may select processor profile.
Public UI-only	Mocked PageData is allowed only for site-local visual/unit work; it is not QA21 integration evidence.
Full E2E	ENV21-E2E starts/builds all 14 site apps + Admin + API against production-shaped routing; used only for release QA, not daily local requirement.

LOCAL SAFETY	Local service names/ports and `BP_SITE_APP_KEY` are convenience metadata only. They never replace real Host-based public SiteContext or selected-site Admin authorization in integration/E2E tests.

12. Immutable Build and Artifact Contract

Artifact	Store/class	Locked contents
REL24-MANIFEST	release-manifest.json	Git SHA, build ID, pnpm-lock.yaml SHA-256, uv.lock SHA-256, OpenAPI SHA-256, migration bundle digest, image digests, static asset digests, affected graph, source document baseline.
OCI images	ECR	One digest per affected site/Admin/API/Worker/migration image.
Static assets	S3 immutable prefix	`web-assets/{app_package}/{git_sha}/...`; immutable and never overwritten.
OpenAPI	contract artifact	Exact committed BLOG-DOC-15 RC7 YAML hash.
DB bundle	migration artifact	Ordered raw SQL + seed/verification scripts and checksum manifest.
QA evidence	artifact store	QA21 results tied to release manifest/build/environment; contains no secrets.
SBOM/scan	artifact store	SPDX JSON SBOM + vulnerability/secret/image scan reports for built images.

A release candidate promoted from Staging to Production references the same image digests and static/migration/contract artifacts. A code or dependency change after Staging validation creates a new release candidate and invalidates candidate-specific QA evidence according to DOC-22 change propagation.

13. CI/CD Architecture and Pipeline Stages

The production reference pipeline uses AWS CodePipeline + CodeBuild with source repository integration and AWS IAM service roles. Pipeline definitions/buildspecs live under `infrastructure/templates/ci/` and AWS resource definitions under `infrastructure/templates/aws/cloudformation/`. No long-lived cloud credential is stored in repository or pipeline variables.

Stage	Name	Required behavior
CI24-01	Source integrity	Checkout exact commit; verify clean ref/base SHA; verify/install frozen `pnpm-lock.yaml` and `uv.lock`; reject package-local or floating dependency authority.
CI24-02	Static architecture	Run Prettier/ESLint/TypeScript/dependency-cruiser plus Ruff/mypy/Import Linter, YAML parse and secret scan.
CI24-03	Contract regeneration	Generate TypeScript OpenAPI types; compare FastAPI runtime registry/derived OpenAPI and frontend clients to DOC-15 RC7; fail on drift.
CI24-04	Affected graph	Produce machine-readable affected.json from Turborepo graph plus declared Python bridge/reverse-dependency inputs (`pyproject.toml`, `uv.lock`, `packages/python/**`).
CI24-05	Unit/integration	Run affected JS/TS tests and Python pytest/integration tests plus repository-wide mandatory regression guards.
CI24-06	Database verification	Start disposable PostgreSQL; apply all migrations from zero; apply allowed seeds; verify 33 tables/20 FORCE RLS/constraints/69 permissions.
CI24-07	Build	Build affected deployable OCI images/static assets; produce migration image when database bundle changes.
CI24-08	Supply-chain checks	SBOM, image vulnerability scan, image signature/attestation, no mutable tags as deploy identity.
CI24-09	Publish	Push artifacts/images once; generate REL24-MANIFEST.
CI24-10	Stage deploy	Deploy same digests to Staging; run QA21 release gates/evidence.
CI24-11	Production approval	Manual protected approval after all required gates + operational preflight.
CI24-12	Production deploy	Promote same digests; rollout, smoke, monitor, evidence record or rollback.

14. Affected Dependency Graph and Selective Builds

Change class	Affected result
apps/sites/site-XX/** only	That Next.js site app + local tests; no other site image.
apps/admin/** only	Admin + its tests.
apps/api/** only	Python API image + pytest/integration/contract tests; site images are not rebuilt unless a wire/shared TS dependency changed.
apps/worker/** only	Python Worker image + worker tests; API/site images are not rebuilt unless a shared Python/wire dependency changed.
packages/ts/api-contract or contracts/http	API contract + both TypeScript clients + API + Admin + all 14 site apps + contract QA.
packages/python/bp-domain|bp-ports|bp-application/**	API + Worker + dependent Python tests; frontends only if API contract/public client changes.
packages/python/bp-adapters-*/**	API and/or Worker consuming the adapter + Python integration tests.
database/migrations|seeds|rls	Migration image + DB/ISO/REG tests + API/Worker integration; public frontends not automatically rebuilt.

Change class	Affected result
root ecosystem/task configuration	Turborepo re-evaluates declared task nodes. pnpm/TS and uv/Python lock/config changes invalidate the appropriate ecosystem graph; turbo.json changes re-evaluate the full task graph.
infrastructure/images/<class>.Dockerfile	Only images using that Dockerfile, plus container/security checks.
deployment/Kubernetes/CI template only	No application rebuild if image contents unchanged; deployment validation still required.
pyproject.toml | uv.lock | Ruff/mypy/Import-Linter config	API + Worker + Python packages/tests are affected; public-site rebuild is not implied by a Python-only dependency change.
package.json | pnpm-lock.yaml | pnpm-workspace.yaml | TS lint/type config	JS/TS workspaces are affected according to Turborepo edges; package-local JS lockfiles remain prohibited.

FAN-OUT RULE	Affected execution is an optimization, never a permission to skip global invariants. Cheap repository-wide contract/count/import/security checks still run on every candidate even when only one public site is rebuilt.

15. Repository-Wide Mandatory CI Gates

Gate	Repository-wide assertion
QA21-REG-001	Estate counts: 14 current site apps, 19 modules, 36 Admin pages, 14 public Page IDs, 161 capabilities, 69 permissions.
QA21-REG-002	DOC-08 DB/seed counts and FORCE RLS verification.
QA21-REG-003	12 ArticleBody types end-to-end; no raw/custom HTML escape.
QA21-REG-004	No site-to-site import/dependency edge.
QA21-REG-005	DOC-15 RC7 PDF/YAML/route/client contract sync: 112 ops / 85 paths / 217 schemas / 1,297 refs.
QA21-REG-006	DOC-16 golden SEO/search/sitemap/redirect vectors where relevant; always on release candidate.
QA21-REG-007	DOC-17 16-key derivative and private-original/native_webp baseline registry.
QA21-REG-008	DOC-18 6 queues / 11 jobs / 7 retry profiles / Redis prefixes/dedupe state.
QA21-REG-009	11 API15-AUTH operations and DOC-19 security constants/mechanics alignment.
QA21-REG-010	Release diff introduces no unauthorized feature/permission/DB/API/Page/context/media/search behavior.

- Secret scanning, YAML parsing, dependency vulnerability policy, Dockerfile policy and generated-file drift are additional deployment checks; they do not replace QA21 IDs.
- A repository-wide gate failure stops artifact publication even if the affected application build itself succeeds.

16. OpenAPI and Generated-Contract Gate

The committed `contracts/http/BLOG-DOC-15_openapi_v1.0-RC7.yaml` is hashed before build. CI regenerates TypeScript wire types/clients, validates the FastAPI runtime route registry and derived framework OpenAPI against the canonical contract, and requires zero contract drift. The API image label/release manifest records the OpenAPI hash. Production cannot run a route/schema/status set different from the staged RC7 contract.

NO CODE-FIRST DRIFT	FastAPI-generated OpenAPI (`/openapi.json`) is derived/diagnostic only. CI compares it and the runtime route registry to the committed DOC-15 RC7 YAML; it cannot become source authority or overwrite the contract. Any intended HTTP change first revises DOC-15/OpenAPI.

17. Database Migration Authority and Ordering

DOC-08 remains the physical schema authority. DOC-24 only freezes how approved SQL reaches an environment. Migration files are ordered raw SQL under `database/migrations/`; filename ordering begins with the frozen DOC-08 DB-MIG identifier/sequence and includes a readable suffix. Migrations are immutable after they have been applied to Staging or Production.

```
database/migrations/
  0001_DB-MIG-001__bootstrap-global.sql
  0002_DB-MIG-002__site-owned-core.sql
  ...
  <sequence>_<DB-MIG-ID>__<kebab-purpose>.sql

database/seeds/
  permissions.sql
  initial-sites.sql
  development-only.sql
  isolation-fixtures.sql

database/verification/
  verify-schema.sql
  verify-rls.sql
  verify-seeds.sql
```

- Checksums for every migration file are stored in the release migration manifest; an already-applied file checksum changing is release-blocking.
- Migration runner acquires a PostgreSQL advisory deployment lock before any schema step so two releases cannot migrate concurrently.
- Each migration executes in a transaction unless the SQL operation explicitly requires non-transactional execution; such a file must be separately marked/reviewed and still participates in the migration lock/checksum ledger.
- Application pods never own migration credentials. IMG24-MIGRATE uses a short-lived deployment identity with only the privileges required for approved schema/seed operations.
- Production migration runner sets a bounded `lock_timeout` and `statement_timeout`; a timeout fails the release instead of waiting indefinitely behind live traffic.

18. Migration File and Execution Contract

Step	Phase	Required behavior
MIG24-01	Preflight	Verify target environment, current schema version/checksums, no concurrent migrator, backup/PITR health and release manifest.
MIG24-02	Lock	Acquire one platform migration advisory lock.
MIG24-03	Plan	List pending files in exact lexical/sequence order; reject gaps, duplicates, checksum mutation or unknown migration.
MIG24-04	Apply	Execute one file at a time with explicit transaction mode and safe timeouts.
MIG24-05	Verify	Run schema/constraint/index/RLS/seed invariants immediately after pending set.
MIG24-06	Record	Persist deployment migration ledger in deployment evidence, not a new product table; provider/native migration metadata may be used outside DOC-08 schema.
MIG24-07	Release lock	Release only after success/failure state and evidence are written.

19. Production Migration Safety Classes

Class	Meaning	Safety rule	Production disposition
M24-EXPAND	Additive/backward compatible	New nullable column/index/table/constraint behavior only when already authorized by DOC-08 revision; previous production code remains functional.	May run before application rollout after preflight.
M24-BACKFILL	Data transformation using existing approved schema	Idempotent, restartable, bounded batches; does not redefine business meaning.	Runs as explicit deployment job; monitor/stop/restart safely.
M24-CONTRACT	Drop/rename/tighten/remove compatibility	Potentially breaks old code/rollback.	Never same release as first consumer switch; requires later approved release, dependency proof, staging rehearsal and manual production window.
M24-EMERGENCY-FIX	Forward-only corrective SQL	Only to repair failed authorized migration/data issue without inventing schema semantics.	Must become a committed migration immediately; no untracked console-only final state.

NO AUTO DOWN	Production rollback does not automatically execute reverse/down migrations. Schema rollback is allowed only when an explicitly reviewed inverse is proven safe and does not discard data. Otherwise restore application digests and forward-fix the database, or invoke controlled PITR only for a true recovery event.

20. Seed Execution and Environment Rules

Seed class	Allowed tier	Rule
Permission registry seed	DEV/TEST/STG/PROD	Exactly the frozen 69 keys; idempotent verification. Never delete/reassign unknown permissions silently.
Initial 14-site registry seed	Fresh bootstrap only as DOC-08 authorizes	Creates canonical baseline only on an empty approved environment; not rerun as a destructive sync.
Development fixtures	DEV only	Allowed; cannot ship in production migration image execution plan.
Isolation/QA fixtures	TEST/LOAD/E2E disposable datasets	Deterministic QA21 fixtures; prohibited in Production.
Production business data	PROD	Not managed by generic seed scripts. Runtime Admin/API flows remain authority.

21. RLS and Schema Post-Migration Verification

Verification	Pass condition
Schema inventory	Exactly the current DOC-08 table/constraint/index baseline plus only later explicitly approved DOC-08 revisions.
RLS inventory	All site-owned canonical tables have RLS and FORCE RLS as required; normal runtime roles cannot bypass.
Role split	Site-runtime and platform-runtime roles/privileges remain separate; migration owner is not used by application pods.
Tenant variable	Site transaction test proves `SET LOCAL app.site_id` disappears after commit/rollback and pooled connection reuse.
Cross-site probes	Representative and QA21 exhaustive tests prove same ID/slug in different sites cannot cross.
Permission seeds	Exactly 69 current fixed permission keys and prerequisite rules.
OpenAPI/DB drift	No migration has added a wire-visible field/behavior without matching upstream authority.

Failure after migration but before application rollout leaves the release blocked. The team does not continue deployment simply because the SQL completed. The failed environment is assessed under Section 36 and database recovery rules.

22. Staging Deployment and QA21 Release Gates

Gate	Suite	Exit condition
G21-0	REG + API/static	No frozen-count/OpenAPI/import drift.
G21-1	ISO + DB	100% PASS; any isolation leak stops release.
G21-2	PERM + SEC	100% PASS; no auth/secret/privacy bypass.
G21-3	ADM + PUB + CONC + LIFE + MED + SEO + WRK	100% PASS across required matrices.
G21-4	A11Y	All frozen WCAG/semantic/responsive assertions PASS.
G21-5	PERF	DOC-20 numeric load/SLO/fairness gates PASS using ENV21-LOAD evidence.
G21-6	DR	Required PITR/media/derived/full-DR evidence is current and PASS; schema-changing candidate adds migration-specific restore rehearsal when required.
G21-7	Final REG smoke + failure-neighbor reruns	No unresolved RB-BLOCKER/RB-FAIL/BLOCKED/FLAKY release test.

100% MEANS 100%	A Staging deployment is not production-eligible because "most tests passed". DOC-21 requires every applicable canonical QA21 ID and required child parameter instance to PASS with evidence for the exact candidate.

23. Production Provider Topology

DOC-24 selects the following production reference wiring. Provider products implement already-frozen interfaces; they are replaceable only through controlled DOC-24 impact review proving identical upstream semantics.

Concern	Selected provider	Locked use
Compute/orchestration	AWS EKS	Separate Staging and Production clusters/accounts; Kubernetes Deployments/Services/Kustomize overlays.
Registry	Amazon ECR	Immutable OCI images addressed by digest.
Ingress/TLS	AWS Application Load Balancer + ACM + Route 53	Host/path routing, SNI TLS certificates; original Host preserved to site/API.
Static asset CDN	Amazon CloudFront + S3 immutable web-assets bucket	Versioned site/Admin build assets only; no Admin/Preview response caching.
Media origin/CDN	Private Amazon S3 + CloudFront origin access	Implements DOC-17 private original/derivative/CDN rules; public only through approved derivative delivery.
PostgreSQL	Amazon RDS for PostgreSQL 17-compatible service	Primary authoritative DB; PITR/WAL/base-backup policy must satisfy DOC-20.
Control/cache Redis	Amazon ElastiCache Redis-compatible dedicated replication group	DOC-18 application-visible keys/TTL/locks/dedupe only.
Queue broker	Separate Amazon ElastiCache Redis-compatible replication group	Celery/Kombu durable broker plane plus broker-private delayed-work storage; separate endpoint/credentials/memory policy; DOC-18 semantics remain authoritative.
Search	Amazon OpenSearch Service	Derived site-scoped public index; DOC-16 ranking/alias rules remain authoritative.
Security state	Amazon DynamoDB	Separate environment table with TTL + encryption; implements DOC-19/DOC-23 conditional/TTL SecurityStateStore.
Secrets/Transit	External managed Vault-compatible HA service	Runtime secrets, dynamic credentials where configured, Transit/envelope crypto; separate env policies/keys.
Mail	Amazon SES SMTP endpoint via Python SMTP adapter	Invitation/reset delivery only; adapter cannot change DOC-19 token/confidentiality semantics.
Observability	OpenTelemetry Collector -> CloudWatch Logs + managed Prometheus-compatible metrics + trace backend	Must implement DOC-20 fields/retention/alerts; application code remains vendor-neutral.
CI/CD	AWS CodePipeline + CodeBuild	Pipeline and buildspec YAML from repository; IAM roles, no static cloud keys.

24. Ingress, DNS, TLS, CDN and Host Preservation

- Each current public hostname has Route 53 DNS and an ACM certificate attached to the ALB listener through SNI. Host-based rules route each domain to its owning site Service. A later domain alias/canonical change must be routable/certified before activation.
- For every public site host, path `/api/v1/*` routes to the shared API Service; other public routes route to that site app Service. The original HTTP Host is preserved. The API does not infer tenant from Kubernetes Service, site image or package name.
- Admin uses a dedicated Admin hostname/service. Its `/api/v1/*` requests route to the same API; DOC-19 credentialed CORS/Origin/CSRF rules remain exact.
- CloudFront serves immutable fingerprinted site/Admin static build assets from S3. Dynamic PageData/API/Admin/Preview responses are not cached merely because the edge can cache them.
- Media CloudFront origin access can read only the configured private media origin and must enforce DOC-17 safe public derivative/protected-original paths. Private canonical originals are never anonymous.
- Ingress strips untrusted client-supplied forwarding headers and writes trusted proxy metadata. API trust-proxy configuration accepts effective scheme/host only from the internal ALB/proxy network boundary.
- Unknown Host remains neutral/fail-closed according to DOC-09; a site Deployment receiving traffic cannot self-select its SITE key.

25. Kubernetes Workload and Deployment Contract

Workload	Type	Prod min	Max	Rule
api	Deployment + ClusterIP Service	3	12	RollingUpdate maxUnavailable=0; stateless; HPA/capacity tuned by DOC-20 evidence.
admin	Deployment + Service	2	4	Static server; no backend secret; independent rollout.
site-01..site-14	One Deployment + Service per site	1 each	4 each	Independent image/digest and rollout; site-local change touches only owning site.
worker-publish	Worker image profile	2	6	q18.publish + schedule timing priority.
worker-media	Worker image profile	2	10	q18.media only.
worker-search	Worker image profile	2	8	q18.search only.
worker-sitemap	Worker image profile	1	4	q18.sitemap only.
worker-cache	Worker image profile	2	8	q18.cache only.
worker-maintenance	Worker image profile	1	2	q18.maintenance/reconcile; distributed locks prevent overlap.
worker-scheduler	Worker image scheduler profile	1	2	Periodic schedule reconciliation; same worker code image, not a new app.
db-migrate	Kubernetes Job from migration image	0	1 active	One-shot pre-rollout job; never a long-running product service.

Replica numbers are the production deployment baseline, not proof of capacity. G21-5/LT20 evidence may require raising replica/resource ceilings before production approval. Such capacity tuning cannot change worker concurrency correctness, DB pool maxima or tenant fairness.

- All pods use explicit requests/limits; API/site/worker resource changes are versioned deployment changes and load-tested when material.
- Production PodDisruptionBudgets preserve at least one healthy API/Admin replica and do not intentionally evict all instances of a critical worker profile at once.
- NetworkPolicy/security groups allow only required service paths: frontends -> API; API/Worker -> approved data/providers; no frontend -> DB/Redis/broker/search/private origin/security store.
- API pool max remains <=20 per replica; Worker process pool max <=10; deployment autoscaling cannot violate the DOC-20 aggregate <=70% PostgreSQL max_connections budget.

26. Worker Deployment Profiles and Draining

The same IMG24-WORKER digest runs multiple Kubernetes Deployments selected by immutable processor-profile configuration. This is scaling, not a new Worker application or new job registry.

- On SIGTERM, a worker marks itself draining and stops claiming new jobs before shutdown.
- `terminationGracePeriodSeconds` is 120 seconds by default. An in-flight job that cannot finish safely before grace exits without ACK; DOC-18 lease/at-least-once redelivery and current-state fencing recover it.
- Roll worker profiles independently so a media backlog does not require stopping scheduled publish workers.
- Broker queues/delayed jobs are never deleted by deployment. A rollback uses old Worker image digest against the same frozen DOC-18 payload semantics.
- If a future worker payload schema changes, upstream DOC-18 change control must define rolling compatibility before DOC-24 may deploy mixed producers/consumers.

27. Health, Liveness and Readiness Wiring

Target	Operational surface	Required behavior
HTTP app liveness	GET /health/live	Outside /api/v1; 200 only when the process/runtime server and its health loop are responsive.
HTTP app readiness	GET /health/ready	Outside /api/v1; safe component summary only; 200 when core dependencies required to serve the process safely are ready.
API core readiness	/health/ready	Config/secrets loaded + PostgreSQL primary reachable + pool checkout + site/domain registry query. Search/media/Redis failure does not remove all API replicas if unrelated routes can safely degrade.
Auth dependency	route dependency + readiness detail	Security-state/Vault/crypto failure makes auth operations return safe 503 and component status degraded; it does not automatically destroy public read capacity.
Site Next.js app	/health/live + /health/ready	Node/Next.js server, config and build output are ready. API failure is handled as request 503; site pod is not killed merely because API is temporarily degraded.
Worker profile	/health/live + /health/ready	Liveness = loop responsive; readiness = broker + PostgreSQL + job-family dependencies exactly as DOC-20.
Migration Job	exit code/evidence	No health endpoint; success only after migration + verification completed.

Readiness uses DOC-20 anti-flap semantics: 3 consecutive dependency failures before unhealthy removal and 2 consecutive successes before recovery unless a hard failure is explicitly reported. Probe bodies never expose secrets, connection strings, raw exception traces, private hostnames or site content.

28. Release Promotion and Production Preflight

Check	Pass condition
Artifact identity	Release manifest exists; all images/artifacts match Staging-tested digests and valid signatures/checksums.
QA evidence	G21-0..G21-7 satisfied; no BLOCKED/unresolved FLAKY/RB failure. Evidence belongs exact candidate.
Database health	Primary healthy; connection headroom; no migration already running; schema checksum equals expected predecessor.
PITR/backup	WAL/archive lag <=5m; latest successful base backup age <=24h; daily verification PASS. DOC-20's 26h value is alert-only, not a release gate. Schema-impact release has required restore rehearsal/snapshot evidence.
Critical alerts	No active unresolved critical alert that makes rollout unsafe, including audit write failure or backup/PITR failure.
Queue health	Publish queue not beyond timing breach; unexpected DLQ critical conditions resolved. Deployment does not purge queues.
Secrets	Required Vault policies/keys/secret refs present; no expired/Development secret ref.
Ingress/cert	Affected domains have valid TLS/routing and Host preservation tests.
Capacity	Planned replicas/pools satisfy DOC-20 connection/fairness budgets and latest load evidence.
Approval	Protected manual Production approval recorded with release manifest ID.

29. Production Rollout Sequence

#	Phase	Required behavior
1	Acquire release execution lock	One Production pipeline only; reject concurrent deployment.
2	Verify immutable artifacts	Digests, OpenAPI hash, migration checksums, signatures, environment target.
3	Run migration preflight	Backup/PITR/schema predecessor/lock/connection headroom.
4	Apply pending M24-EXPAND migrations	IMG24-MIGRATE Job; stop on any verification failure.
5	Deploy API	Rolling update; maxUnavailable=0; readiness/synthetic API checks.
6	Deploy Worker affected profiles	Drain old; roll one profile at a time; keep queues/delayed jobs.
7	Deploy Admin if affected	Independent digest; auth/CSRF/session smoke.
8	Deploy affected public site apps	One site at a time by affected graph; other site deployments remain untouched.
9	Publish immutable static assets	Already built digests/prefixes; switch manifest references only after upload verified.
10	Run post-deploy smoke	All affected hosts/routes, SiteContext isolation, auth, publish/queue checks as applicable.
11	Observe release window	At least 30 minutes of error/latency/pool/queue/audit/security/edge telemetry; no critical regression.
12	Finalize evidence	Record release status, exact digests, migration set, QA/smoke/telemetry links in DOC-22 evidence system.

SITE INDEPENDENCE	A change confined to SITE-03 presentation deploys only APP-03. It does not redeploy or restart the other thirteen public applications, API, Worker or database unless the affected graph includes a shared dependency or migration.

30. Application Rollback Contract

Target	Rollback action
Site app	Redeploy previous known-good site image digest; keep current DB if backward-compatible. Other sites unaffected.
Admin	Redeploy previous Admin digest; API contract must remain compatible during rollback window.
API	Roll back to previous API digest only if current database and auth/security state remain compatible; default coordinate Worker rollback when shared backend package changed.
Worker	Redeploy previous Worker digest/profile. In-flight unacked work redelivers under DOC-18; queues are not cleared.
Static assets	Switch app/manifest to previous immutable asset prefix; never overwrite old prefix.
Configuration	Roll back to previous versioned non-secret manifest/secret reference. Never restore an old secret value solely to match code if security rotation has revoked it.

Rollback trigger includes stop-the-line QA failure, tenant/security anomaly, sustained critical SLO regression, migration/application incompatibility, or operator-confirmed functional corruption. Rollback does not erase evidence; correlation IDs, logs and failed release artifacts remain retained under DOC-19/20/22.

31. Database Rollback Limitations and Forward Recovery

Database state is not treated like an OCI image. Production application rollback is designed to work without reversing ordinary expand migrations. Automated `down` execution is prohibited because a reverse DDL may destroy data created after migration or violate newer constraints.

Situation	Required response
Migration failed before commit	Transaction rollback where supported; release remains blocked; investigate and create corrected committed migration.
Non-transactional index/DDL failed	Inspect exact provider state; cleanup only through reviewed forward corrective migration; do not continue code rollout.
Migration succeeded, app failed	Roll application to previous digest if schema is backward compatible; keep additive schema.
Data backfill partially completed	Stop; rerun idempotent backfill from recorded checkpoint/batch logic; app remains compatible with both old/new representation until complete.
Contract/destructive migration issue	No automatic reverse. Invoke approved forward fix or, if canonical corruption makes recovery necessary, DR/PITR decision under DOC-20 with platform-wide consequences.
Tenant data corruption discovered	Stop release, preserve evidence, determine blast radius. PITR is not casually used for one row/site because physical DB recovery is platform-wide; use authorized logical repair only if upstream rules permit.

PITR IS RECOVERY	PostgreSQL PITR is adopted with RPO <=5 minutes and RTO <=60 minutes, but it is a disaster-recovery control, not the normal rollback button for a bad application deploy. Any restore occurs to an isolated target and must pass RLS/site/audit integrity before traffic.

32. Expand / Backfill / Contract Migration Strategy

```
Release N      : EXPAND schema (old code still works)
Release N        : deploy code that can read old + write compatible new form
Optional job     : idempotent BACKFILL in bounded batches
Observation    : verify QA21 + telemetry + no old-reader dependency
Later Release  : switch reads fully to new form while old structure still present
Contract Release: remove old structure only after dependency scan, staging rehearsal,
                  current backup/PITR evidence and manual approval
```

- Contract migration is never bundled into the first release that introduces the replacement structure.
- Before a contract migration, repository search/dependency checks prove no previous production-compatible image expected for rollback still requires the removed field/object.
- If a schema change would alter the frozen 33-table/column/constraint model, DOC-08 must already have been revised and frozen before DOC-24 may schedule it.
- Long backfills are separated from API pod startup and do not hold an application request transaction open.

33. Configuration and Infrastructure Drift Control

- Production Kubernetes rendered manifests, CloudFormation stacks, Route 53/ALB/CloudFront configuration classes and Vault policy references are versioned deployment sources. Console hot-fixes are emergency-only and must be reconciled back to source before the next release.
- Nightly/continuous drift detection compares provider state to the approved stack/manifests. Security-group/public-bucket/IAM/route/certificate drift is high priority.
- Drift detection never auto-deletes unknown production resources. It reports and blocks promotion until reviewed.
- A secret VALUE change is not source drift; a secret REFERENCE/policy/key-name change is versioned deployment config.
- Environment-specific replica counts/endpoints/resource IDs may vary through overlays, but no overlay may change API route semantics, permission mapping, Page IDs, worker queue/job names, Redis key grammar or database tenant rules.

34. Observability and Alert Wiring

Signal	Wiring	Rule
Logs	Structured JSON + OpenTelemetry -> collector -> CloudWatch Logs	30-day application operational retention; Python and Node emit the same DOC-20 safe fields; security/audit retention follows DOC-19/20.
Metrics	Prometheus-compatible exporter -> managed metrics backend	Implement DOC-20 metric names/cardinality; environment/service/site dimensions only where allowed.
Traces	OTel -> trace backend	Sampling exactly follows DOC-20 policy; 100% of 5xx/timeouts/backup/restore paths where required.
Alerts	Managed alert rules + SNS/on-call destination	Implement ALT20-01..20 thresholds exactly; deployment cannot weaken threshold to silence a failing release.
Release marker	REL24 manifest ID emitted as telemetry attribute	Correlates latency/error/queue/DB changes to exact rollout; not a tenant authority.
Audit	Existing audit_events path	Deployment does not create a second business audit table; security/audit failures remain stop-the-line.

Production deployment emits a release-start and release-complete operational marker plus image digest/release SHA. It does not log secret values, raw session tokens, Preview credentials, article bodies, object-store private keys or client idempotency values.

35. Backup/PITR and DR Operational Integration

Asset	Provider mechanism	Contract
PostgreSQL	RDS automated backups + continuous WAL/PITR	Base-backup cadence <=24h; Production release-preflight age <=24h; recoverable window 35 days; archive lag <=5m; RPO <=5m; RTO <=60m. DOC-20 26h is alert-only.
Private media originals	S3 versioning/replication or equivalent provider durability	35-day recoverable window; RPO <=15m; RTO <=2h; deleted media backup residual never public.
Redis/control	Provider HA/persistence for availability	Not canonical backup; reconstruct per DOC-18.
Queue broker	Durable Redis broker persistence/replication	Operational durability; reconciliation repairs inferable gaps; not database backup.
OpenSearch	Optional snapshots for speed	Derived; may rebuild from PostgreSQL using DOC-16/18.
Sitemap/variants	Rebuildable	No canonical backup requirement.
Security state	Provider-native encrypted backup/recovery per DOC-19 incident model	Recovery may force reauthentication; never restore revoked/expired secret/token blindly.
DR evidence	Isolated ENV21-DR target	Quarterly restore drills and six-month full-platform game day per DOC-20; evidence retained at least 365 days.

36. Failure Handling and Interrupted Release State Machine

Failure	Required response
Preflight fails	No mutation/deployment. Candidate remains staged; fix cause and rerun required gates.
Migration fails	Stop before new application rollout. Preserve DB/migration logs; transactional rollback/forward correction only.
API rollout readiness fails	Pause rollout; old pods remain serving because maxUnavailable=0; roll back new ReplicaSet if not corrected.
One site rollout fails	Roll back only that site digest; do not touch other sites unless shared dependency issue is proven.
Worker rollout fails	Restore prior worker profile digest; unacked work redelivers; do not clear queue/DLQ.
Critical tenant/security alert	Stop deployment immediately; invoke DOC-19 incident containment and preserve evidence.
Performance regression	If DOC-20 critical threshold/SLO gate fails, rollback affected digests or scale only within frozen pool/fairness limits; scaling cannot mask correctness failure.
Edge/certificate failure	Do not activate new domain/route; keep last known-good routing. Never bypass TLS/Host validation.
Post-deploy smoke fails	Release not finalized; rollback affected application(s) and reopen release evidence.
CI runner interruption after artifact publish	Artifacts remain immutable but are not automatically production-eligible; rerun stage/evidence from exact digests.

37. Evidence, Traceability and Change Propagation

Evidence ID	Required contents
EV24-BUILD	Git SHA, pnpm-lock.yaml hash, uv.lock hash, affected graph, build logs, image/static/migration digests, SBOM/signature.
EV24-CONTRACT	OpenAPI hash + generated diff + route/client contract check.
EV24-DB	Migration plan/checksums, apply log, schema/RLS/seed verification, backup/PITR preflight.
EV24-STAGE	Rendered Staging manifests + deployed digests + G21 evidence links.
EV24-PERF	ENV21-LOAD run ID, LT20 scenario outputs and telemetry window.
EV24-DR	Restore target, backup identifiers, measured RPO/RTO, RLS/isolation result.
EV24-PROD	Approval, rendered Production manifest, rollout events, smoke, 30-minute monitor, final status.
EV24-ROLLBACK	Trigger, decision, previous/current digests, DB state, verification and incident/reference if applicable.

Every evidence artifact is attached to the DOC-22 MX22/release implementation row using exact release manifest/build/environment identity. Changing source, migration, config or image digest after evidence collection invalidates the affected evidence by DOC-22 propagation rules. DOC-24 itself does not mark implementation VERIFIED.

38. DOC-01 through DOC-23 Consistency Audit

Authority	Result	Audit conclusion
DOC-01..04	PASS	One shared platform, 14 independent sites, trusted SiteContext/fail-closed isolation preserved through host routing, workload config and data services.
DOC-05..14	PASS	No Page ID/flow/component/data/route/frontend ownership change; deployment keeps 14 site apps independent.
DOC-15 RC7	PASS	112-operation contract is unchanged; health probes are explicitly outside /api/v1; same OpenAPI artifact promoted.
DOC-16 RC4	PASS	CDN/edge/search deployment does not change ranking/canonical/sitemap/redirect semantics.
DOC-17 RC2	PASS	S3/CloudFront wiring keeps canonical originals private and only generated sanitized derivatives public.
DOC-18 RC2	PASS	Separate control/broker Redis, Celery execution mapping, six queue profiles, broker-private delayed scheduling and restart behavior preserve all six queues/eleven jobs/retry/dedupe/lease/DLQ contracts.
DOC-19 RC2	PASS	Vault/security-state/workload identity/edge rules preserve auth/session/MFA/CSRF/CORS/secrets; no prod default credentials.
DOC-20 RC1	PASS	Deployment implements health/SLO/alert/pool/PITR/RPO/RTO/restore requirements without weakening thresholds.
DOC-21 RC1	PASS	Pipeline executes G21-0..7 and preserves 100% applicable PASS; affected builds do not skip repository-wide release guards.
DOC-22 RC1	PASS	Release artifacts/evidence remain NOT_VERIFIED until candidate proof exists; immutable IDs support change propagation.
DOC-23 RC2	PASS	Exact mixed-language Hybrid Monorepo consumed: 14 Next.js sites, React/Vite Admin, Python/FastAPI API, Python/Celery Worker, pnpm+uv locks and revised dependency boundaries. Superseded RC1 runtime references are removed.

AUDIT RESULT	The RC1 cross-document blocker was genuine and is resolved in RC2 by synchronizing every deployment/runtime reference with DOC-23 RC2. No remaining blocking contradiction or missing upstream dependency was identified for DOC-24 scope. DOC-20 health/SLO/DR and DOC-21 release gates remain unchanged.

39. Comprehensive Acceptance Criteria

ID	Release-blocking deployment assertion
AC24-001	Exactly four promotion tiers DEV/TEST/STAGING/PRODUCTION are defined, while QA21 UNIT/INT/E2E/LOAD/DR remain evidence classes rather than hidden product environments.
AC24-002	Staging and Production use isolated cloud accounts/VPCs/clusters/data services and no environment credential authenticates to the other environment by default.
AC24-003	Production data/secrets are not normal Development/Test/Staging seed material.
AC24-004	All runtime non-secret configuration uses validated BP_* inputs and is parsed once before readiness.
AC24-005	No application module reads the process environment directly outside the approved config loader (`process.env` in Node or `os.environ`/equivalent in Python).
AC24-006	Missing required Production configuration fails startup/readiness rather than using Development defaults.
AC24-007	No browser bundle receives DB, Redis, broker, OpenSearch, S3-private, Vault or security-state credentials/endpoints requiring secrecy.

BLOG-DOC-24 | ENVIRONMENT, DOCKER, CI/CD, MIGRATION, DEPLOYMENT & ROLLBACK CONTRACT

ID	Release-blocking deployment assertion
AC24-008	Vault/workload-identity secret injection leaves no long-lived cloud secret in source/image/ConfigMap/build log.
AC24-009	Site-runtime and platform-runtime PostgreSQL credentials remain distinct.
AC24-010	Control/cache Redis and queue-broker Redis remain separate endpoints/credentials in Staging/Production.
AC24-011	Every production OCI image runs as non-root and contains no .env/Git/test-secret/package-cache material.
AC24-012	Runtime images are addressed by digest and no deployment references `latest` as release identity.
AC24-013	Application/container startup performs no CREATE/ALTER/schema synchronization.
AC24-014	Admin image contains only public static runtime config and no server secret.
AC24-015	Migration image is an operational artifact and does not become an eighteenth product application.
AC24-016	All new release-critical YAML uses .yaml, two-space indentation, UTF-8/LF, no literal secrets and validated rendered output.
AC24-017	Production Kubernetes YAML pins image digests and does not rely on mutable tags.
AC24-018	Local compose supplies disposable dependencies without requiring all 14 public application servers.
AC24-019	A developer can run one chosen public site + API with core dependencies while the other 13 site apps remain stopped.
AC24-020	Local dummy Vault/security-state/SMTP services cannot contain production credentials or send production reset/invitation email.
AC24-021	One release manifest binds Git SHA, OpenAPI hash, migration checksums, image/static digests and affected graph.
AC24-022	Staging and Production deploy the identical built image/static/migration artifact digests; any rebuild creates a new candidate.
AC24-023	Site-local code change affects only its site build/tests unless a shared dependency edge changes.
AC24-024	Root task-orchestration changes re-evaluate all task nodes; pnpm/TypeScript and uv/Python dependency-policy changes invalidate the appropriate ecosystem graph without pretending there is one universal lockfile.
AC24-025	OpenAPI/API-contract change affects API, clients, Admin and all public site apps as dictated by dependency graph.
AC24-026	Python domain/application changes affect API and Worker but do not force a public-site rebuild absent a DOC-15/shared TypeScript contract change.
AC24-027	Database migration change triggers DB/ISO/REG/API/Worker integration validation and does not automatically rebuild frontends.
AC24-028	Repository-wide QA21-REG-001..010/static security/import checks run even for a one-site affected build.
AC24-029	CI installs with both authoritative ecosystem locks (`pnpm-lock.yaml` and `uv.lock`) frozen and rejects package-local lockfiles/floating dependency authority.
AC24-030	CI regenerates OpenAPI types and fails if generated output or route/client contract differs from DOC-15 RC7.
AC24-031	CI verifies 112 operations/85 paths/217 schemas/1,297 refs for the RC7 baseline.
AC24-032	CI secret scanning fails on committed credentials/private keys/tokens.
AC24-033	CI builds/publishes artifacts only after static/contract/unit/integration/database gates pass.
AC24-034	Built images produce SBOM and vulnerability/signature evidence before promotion.
AC24-035	Database migration files are ordered raw SQL under database/migrations and applied checksums become immutable.
AC24-036	One PostgreSQL advisory migration lock prevents concurrent migration executions.
AC24-037	Migration preflight rejects wrong predecessor checksum, duplicate/gapped sequence, unhealthy PITR/backup or concurrent migrator.
AC24-038	Each migration is transactional unless explicitly marked/reviewed for a required nontransactional operation.
AC24-039	Migration execution uses bounded lock/statement timeouts and timeout stops the release.

BLOG-DOC-24 | ENVIRONMENT, DOCKER, CI/CD, MIGRATION, DEPLOYMENT & ROLLBACK CONTRACT

ID	Release-blocking deployment assertion
AC24-040	Post-migration verification proves current DOC-08 schema/RLS/role/seed invariants before app rollout.
AC24-041	No migration may add/change an upstream database semantic unless DOC-08 has already been revised/frozen.
AC24-042	Permission seed remains exactly the current 69 keys and is idempotently verified.
AC24-043	Development/isolation fixture seeds never execute in Production.
AC24-044	Initial site seed is bootstrap-only and not a destructive sync against existing Production sites.
AC24-045	M24-EXPAND changes remain backward compatible with the previous production application.
AC24-046	M24-BACKFILL jobs are idempotent/restartable/bounded and are not application startup work.
AC24-047	M24-CONTRACT change is not shipped in the same release as its initial replacement structure/consumer change.
AC24-048	Production auto-down migration is prohibited.
AC24-049	Application rollback normally leaves additive schema in place and restores previous image digest.
AC24-050	PITR is treated as recovery, not routine application rollback.
AC24-051	Staging deploys the exact candidate and executes G21-0..G21-7 without weakening DOC-21 expected results.
AC24-052	Every applicable QA21 test/child instance is PASS before Production; percentage pass is rejected.
AC24-053	ENV21-LOAD evidence uses DOC-20 numeric thresholds exactly.
AC24-054	Current DR/PITR/media/full-game-day evidence satisfies DOC-20 cadence before release gate G21-6 passes.
AC24-055	Production provider wiring uses one EKS architecture with separate site/Admin/API/Worker workloads and does not clone backend services per site.
AC24-056	Each of the 14 site apps has an independent Kubernetes Deployment/Service/image digest.
AC24-057	Host-based ingress routes each public hostname to its owning site and /api/v1 to shared API while preserving trusted Host.
AC24-058	Unknown Host cannot fall back to a site based on service/package metadata.
AC24-059	Admin has a dedicated hostname and cannot import or route through a public site presentation runtime.
AC24-060	CloudFront/static asset caching never overrides Admin/Preview no-store rules.
AC24-061	Media origin remains private; CloudFront exposes only DOC-17-approved derivative delivery and never anonymous canonical originals.
AC24-062	API autoscaling cannot violate per-replica pool max <=20 or aggregate <=70% max_connections budget.
AC24-063	Worker deployment autoscaling cannot violate per-process pool max <=10, DOC-18 concurrency/fairness/lease semantics or queue correctness.
AC24-064	Worker profile rollout uses the same Worker code image and does not create new job types/apps.
AC24-065	Worker termination stops new claims; unfinished unacked work is recovered by DOC-18 at-least-once semantics rather than silently ACKed.
AC24-066	Broker queues/delayed jobs survive deployment and are not purged by rollout/rollback.
AC24-067	/health/live and /health/ready are operational routes outside /api/v1 and expose no secrets/site data.
AC24-068	API readiness requires safe PostgreSQL core access but does not make unrelated dependency degradation remove all public capacity.
AC24-069	Worker readiness matches DOC-20 family-specific dependencies.
AC24-070	Readiness anti-flap uses 3 consecutive failures and 2 successes unless hard failure.
AC24-071	Production preflight verifies exact artifact identity, QA evidence, DB health, PITR/WAL health, latest successful base-backup age <=24h, critical alerts, queue health, secrets, TLS/Host and capacity.

ID	Release-blocking deployment assertion
AC24-072	Production pipeline prevents concurrent releases.
AC24-073	Production rollout order is migrations -> API -> affected Worker profiles -> Admin -> affected sites -> smoke/observe/evidence.
AC24-074	A one-site rollout failure can roll back that site without redeploying the other thirteen sites.
AC24-075	API/Worker shared-backend rollback is coordinated when compatibility requires it.
AC24-076	Rollback never clears control Redis, Celery broker/delayed work, search or media derived state merely to hide a deployment problem.
AC24-077	Old immutable static asset prefixes remain available for application rollback and are never overwritten.
AC24-078	Critical tenant leak/auth bypass/secret/audit/backup failure stops deployment and invokes security/recovery handling.
AC24-079	Infrastructure/manifest drift is detected and blocks promotion until reviewed; automatic destructive drift reconciliation is prohibited.
AC24-080	Operational alert rules implement DOC-20 ALT20 thresholds without weakening them to make a release pass.
AC24-081	Production release emits safe release markers/digests for observability correlation and no secret/raw content.
AC24-082	PostgreSQL PITR remains continuous with base-backup cadence <=24h and Production preflight age <=24h, 35-day window, archive lag <=5m, RPO <=5m and RTO <=60m.
AC24-083	Private media recovery remains 35-day protected, RPO <=15m and RTO <=2h, with no deleted-original public resurrection.
AC24-084	Derived Redis/search/sitemap/media variants are rebuildable and are not misclassified as canonical backup truth.
AC24-085	DR restores occur in isolated ENV21-DR target and pass RLS/site/audit integrity before any traffic decision.
AC24-086	Every release evidence record references exact candidate build/revision/environment and DOC-22 trace row.
AC24-087	A changed source/config/migration/image digest invalidates affected prior evidence and creates a new candidate where required.
AC24-088	DOC-24 does not mark implementation VERIFIED because this PDF exists; evidence remains required.
AC24-089	No deployment provider/tool introduces a new feature, permission, capability, Page ID, API operation, DB object, job type, media variant or search rule.
AC24-090	Future site deployment creates one new site app/route/certificate/config set and does not clone API/Worker/PostgreSQL/Redis/OpenSearch services.
AC24-091	Domain activation is blocked until DNS/TLS/ingress support exists; infrastructure cannot silently bypass DOC-09 canonical/alias behavior.
AC24-092	Cross-document audit reports zero unresolved blocking contradiction before formal DOC-24 freeze.
AC24-093	IMG24-SITE-* runs Node 24 + Next.js 16.3, while IMG24-API and IMG24-WORKER run Python 3.13 with FastAPI/Celery respectively; no superseded public/API/worker runtime remains normative.
AC24-094	CI and image builds use `pnpm-lock.yaml` for JS/TS and `uv.lock` for Python; neither lockfile is omitted or treated as a generated substitute for the other.
AC24-095	FastAPI-generated OpenAPI is diagnostic only and the runtime registry/derived schema remains contract-equal to DOC-15 RC7.
AC24-096	Celery/Redis deployment preserves DOC-18 six queues, eleven jobs, retry/dedupe/lease/DLQ/reconciliation rules and the DOC-23 broker-private far-future delay mechanism.

40. Freeze-Readiness Checklist

Freeze ID	Required condition
FZ24-01	DEV/TEST/STAGING/PRODUCTION promotion model and QA auxiliary environments accepted.
FZ24-02	Environment isolation/data/credential boundaries accepted.
FZ24-03	BP_* typed configuration + Vault/workload identity secret injection accepted.

Freeze ID	Required condition
FZ24-04	Five image classes and non-root/multi-stage/digest rules accepted.
FZ24-05	`.yaml` standard, infrastructure/templates layout and rendered-manifest evidence accepted.
FZ24-06	Local Compose dependency profiles and selective one-site startup accepted.
FZ24-07	Immutable build-once/promote-by-digest release manifest accepted.
FZ24-08	CI stage/affected-graph/repository-wide REG gate model accepted.
FZ24-09	DOC-15 RC7 OpenAPI regeneration/route/client sync gate accepted.
FZ24-10	Raw SQL migration/checksum/advisory-lock execution model accepted.
FZ24-11	Expand/backfill/contract and no-auto-down production safety accepted.
FZ24-12	Seed environment restrictions + post-migration RLS verification accepted.
FZ24-13	Staging G21-0..G21-7 production-promotion gate accepted.
FZ24-14	AWS EKS/RDS/Redis/OpenSearch/S3/DynamoDB/Vault/SES provider wiring accepted.
FZ24-15	Independent 14-site deployments + shared API/Worker and host-preserving ingress accepted.
FZ24-16	Operational /health/live and /health/ready behavior accepted.
FZ24-17	Production preflight/rollout/application rollback rules accepted.
FZ24-18	DOC-20 backup/PITR/RPO/RTO and DR integration accepted, including Production release-preflight base-backup age <=24h.
FZ24-19	96 AC24 release-blocking assertions accepted; DOC-01..23 audit has zero blocker.
FZ24-20	Implementation remains NOT_VERIFIED/EVIDENCE_REQUIRED until pipeline/images/manifests/migration/QA evidence is supplied.
FZ24-21	DOC-23 RC2 stack synchronization accepted: Next.js public runtime, Python/FastAPI API and Python/Celery Worker images.
FZ24-22	Mixed `pnpm-lock.yaml` + `uv.lock` build/CI/dependency-boundary model accepted.

REVIEW STATUS	BLOG-DOC-24 v1.0-RC3 is the DOC-23 RC2 synchronized deployment/release review candidate with the DOC-20 backup release-gate correction applied. No cloud environment or pipeline implementation is claimed; formal freeze still requires explicit project approval.

Appendix A - Environment / Configuration Variable Registry

Variable	Consumer	Presence	Meaning	Exposure
BP_ENV	ALL	REQ	development|test|staging|production	Nonsecret
BP_RELEASE_SHA	ALL	REQ	Git commit identity	Nonsecret
BP_SERVICE_NAME	server workloads	REQ	api|worker|admin|site-*	Nonsecret
BP_HTTP_PORT	site/api/health servers	REQ	Container listener port	Nonsecret
BP_LOG_LEVEL	server workloads	REQ	Validated log level	Nonsecret
BP_OTEL_EXPORTER_OTLP_ENDPOINT	server workloads	REQ STG/PROD	Collector endpoint	Nonsecret
BP_SITE_APP_KEY	site-*	REQ	APP/SITE diagnostic identity only	Nonsecret; never tenant authority
BP_API_BASE_PATH	browser apps	REQ	/api/v1	Browser allowlist
BP_API_INTERNAL_ORIGIN	site SSR	OPT	Internal API service origin if used	Server-only nonsecret; Host still request-derived
BP_TRUSTED_PROXY_MODE	api	REQ STG/PROD	ALB/internal proxy trust policy identifier	Nonsecret
BP_POSTGRES_HOST	api/worker/migrate	REQ	RDS/PG host	Server-only nonsecret
BP_POSTGRES_PORT	api/worker/migrate	REQ	5432	Server-only nonsecret
BP_POSTGRES_DATABASE	api/worker/migrate	REQ	Database name	Server-only nonsecret
BP_POSTGRES_SSL_MODE	api/worker/migrate	REQ	verify-full in STG/PROD	Server-only nonsecret
BP_POSTGRES_POOL_MAX	api/worker	REQ	<=20 API; <=10 Worker	Server-only nonsecret
BP_POSTGRES_SITE_CREDENTIAL_FILE	api/worker	REQ	/run/secrets/bp/postgres-site	Secret reference path
BP_POSTGRES_PLATFORM_CREDENTIAL_FILE	api/worker	REQ where GLOBAL path	/run/secrets/bp/postgres-platform	Secret reference path
BP_REDIS_CONTROL_ENDPOINT	api/worker	REQ	Control/cache replication group	Server-only
BP_REDIS_CONTROL_PASSWORD_FILE	api/worker	REQ if password auth	Mounted secret file	Secret reference
BP_REDIS_BROKER_ENDPOINT	api/worker	REQ	Dedicated broker replication group	Server-only
BP_REDIS_BROKER_PASSWORD_FILE	api/worker	REQ if password auth	Mounted secret file	Secret reference
BP_OPENSEARCH_ENDPOINT	api/worker	REQ	Environment OpenSearch endpoint	Server-only
BP_MEDIA_BUCKET	api/worker	REQ	Private media origin bucket	Server-only
BP_MEDIA_STAGING_PREFIX	api/worker	REQ	Private staging prefix	Server-only
BP_MEDIA_CDN_ORIGIN	api/public projection	REQ	HTTPS public derivative origin	Safe output config
BP_SECURITY_STATE_TABLE	api	REQ	Environment DynamoDB table	Server-only nonsecret
BP_VAULT_ADDR	api/worker/migrate as needed	REQ	Vault-compatible endpoint	Server-only nonsecret
BP_VAULT_ROLE	workload	REQ	Environment/workload auth role	Nonsecret policy identity
BP_SMTP_HOST	api mail adapter	REQ	SES SMTP endpoint	Server-only
BP_SMTP_PORT	api mail adapter	REQ	TLS SMTP port	Server-only
BP_SMTP_CREDENTIAL_FILE	api mail adapter	REQ	Mounted secret	Secret reference
BP_WORKER_PROFILE	worker	REQ	publish|media|search|sitemap|cache|maintenance|scheduler	Nonsecret; does not add job type

This registry freezes deployment names for the current stack classes, not business configuration. Site/domain registry, permissions, SEO, platform settings and content never move into environment variables. Secret values themselves are not environment variables in Staging/Production; *_FILE entries reference runtime-mounted secret material.

Appendix B - Container / Image Registry

Image ID	Source	Runtime target	Registry class	Rule
IMG24-SITE-01..14	apps/sites/site-*	site-specific	ECR `bp/site-*`	Independent Node 24 / Next.js 16.3 digest per site; future sites add a new repository/image under the same rule.
IMG24-ADMIN	apps/admin	admin	ECR `bp/admin`	Static build server; no secret.
IMG24-API	apps/api	api	ECR `bp/api`	Python 3.13 / FastAPI 112-operation API runtime; canonical DOC-15 RC7 contract unchanged.
IMG24-WORKER	apps/worker	worker profiles	ECR `bp/worker`	Python 3.13 / Celery 5.6 runtime; same digest across six queue profiles + scheduler deployment path.
IMG24-MIGRATE	database + tooling/db	one-shot Job	ECR `bp/migrate`	Python 3.13 one-shot raw SQL/seeds/verification runner using frozen uv environment; not business runtime.

Appendix C - CI Gate Matrix

Trigger	Mandatory gates
PR - every change	Frozen pnpm+uv lock install; JS/TS lint/type/dependency-cruiser; Python Ruff/mypy/Import Linter; YAML parse; secret scan; OpenAPI generation; affected tests/build dry-run; QA21-REG-001..005,008..010 as applicable and cheap global guards.
PR - DB change	Plus fresh-db migration from zero, seed verify, 33-table/20-RLS/69-permission checks, cross-site RLS tests.
PR - public site	Build affected site only; site-local tests; no other site build unless shared package changed.
PR - contracts/shared client	Build/test API clients + Admin + all 14 sites + API contract checks.
Main release build	All PR gates for candidate + build/push affected immutable artifacts + SBOM/signatures + release manifest.
Staging	Deploy same digests; G21-0..4; candidate-specific functional/security/isolation/E2E evidence.
Load	G21-5 using ENV21-LOAD and exact LT20 thresholds.
Recovery	G21-6 current DR evidence; run migration-specific restore rehearsal when required.
Production	Manual approval + preflight + same digest deployment + smoke + monitor + final evidence.

Appendix D - Migration / Release Decision Matrix

Change	Release path	Rollback/recovery
No DB change	Deploy affected app(s) after normal stage gates	Rollback app digest
Additive schema only	M24-EXPAND before app; verify; deploy compatible code	Rollback app; leave additive schema
Large existing-data transform	EXPAND -> deploy compatible code -> explicit M24-BACKFILL -> verify	Stop/restart backfill; app remains dual-compatible
Drop/rename/tighten	Separate M24-CONTRACT release only after old-reader proof and rehearsal	No automatic reverse; forward fix/PITR only if recovery event
Site-only frontend change	Deploy one site image only	Rollback one site image only
Worker-only bootstrap change	Deploy affected worker profiles only	Rollback worker digest; unacked jobs redeliver
OpenAPI change	Requires upstream DOC-15 revision + all affected clients/API/stage QA	Rollback coordinated compatible digests; no runtime schema drift
Critical security incident	Stop release; DOC-19 containment/recovery	Security incident decision outranks deployment completion

Appendix E - Production Deployment Order

```
1. Acquire production release lock
2. Verify REL24-MANIFEST + image/static/migration/OpenAPI digests
3. Verify QA21 gate evidence + protected approval
4. Verify RDS PITR/base-backup/WAL + DB pool headroom + no critical blocker
5. Run IMG24-MIGRATE pending M24-EXPAND files
6. Run schema/RLS/seed verification
7. Rolling deploy API
8. Rolling deploy affected Worker profiles (drain old consumers)
9. Deploy Admin if affected
10. Deploy affected public site apps one-by-one
11. Verify immutable static asset references/CDN
12. Run host/SiteContext/auth/worker/site smoke checks
13. Observe 30-minute release telemetry window
14. Mark success or execute targeted rollback/incident path
15. Attach immutable evidence to DOC-22 traceability record
```

Appendix F - One-Page Developer Execution Reference

Area	Locked answer
ENVIRONMENTS	DEV local, TEST ephemeral CI, STAGING production-shaped nonprod AWS, PRODUCTION separate AWS. LOAD/DR are QA evidence environments, not promotion tiers.
BUILD	Two authoritative ecosystem locks: pnpm-lock.yaml (JS/TS) + uv.lock (Python). Turborepo + Python bridge inputs compute the affected graph. Build once; promote immutable digests; never rebuild for Prod.
SECRETS	BP_* nonsecret config + *_FILE secret references. Vault/workload identity. No secret in source/YAML/image/browser/CI artifacts.
DOCKER	Site/Admin/API/Worker + operational migration image. Multi-stage, non-root, digest pinned, no schema mutation at startup.
LOCAL	Compose dependencies by core/search/media/security/observability profile. Start only selected site + required API/Admin/Worker; other sites can stay stopped.
CI	Affected JS/TS + Python tests/builds plus repository-wide QA21 guards. ESLint/TS/dependency-cruiser and Ruff/mypy/Import Linter run as applicable; OpenAPI is generated/diff checked every candidate.
DB	Raw ordered SQL only. Advisory migration lock. Preflight PITR with base-backup age <=24h (26h alert-only). EXPAND before code. BACKFILL explicit. CONTRACT later. No automatic production down migration.
STAGING	Deploy exact candidate and satisfy G21-0..7. 100% applicable PASS with evidence; BLOCKED/FLAKY is not pass.
PROD	AWS EKS + ECR + ALB/ACM/Route53 + RDS + separate Redis planes + OpenSearch + private S3/CloudFront + DynamoDB + Vault + SES.
ROUTING	Host-based site service routing; `/api/v1` -> shared API; preserve trusted Host; unknown Host fails closed.
WORKERS	One Python/Celery Worker image, six queue-family profiles + scheduler path. Drain on rollout; do not purge broker/delayed work. At-least-once redelivery/fencing remains DOC-18.
HEALTH	`/health/live` and `/health/ready` are operational, outside `/api/v1`; safe dependency status only.
ROLLBACK	Rollback immutable app digest while keeping backward-compatible additive DB state. PITR is DR, not routine rollback.
EVIDENCE	Release manifest, CI, migration, QA, load/DR, staging/prod rollout evidence map to DOC-22; implementation remains NOT_VERIFIED until supplied.